Literature Status for arXiv:math/0210287: the RNP Quotient Question

Run fa_banach_001, agent_lane_13

2026-06-16

Status

This is a `literature_already_answered` packet. It records that the second final question in Kadets–Kalton–Werner, *Remarks on rich subspaces of Banach spaces*, arXiv:math/0210287, was later answered negatively by Kadets–Shepelska–Werner, *Quotients of Banach spaces with the Daugavet property*, arXiv:0806.1815. This packet also retains the earlier local positive partition subcase as included history, not as a separate result packet.

Source Question

The source paper arXiv:math/0210287 ends with two questions of A. Pelczynski. The second asks whether, if X is a subspace of L_1 with the Radon–Nikodym property, L_1/X must have the Daugavet property.

Supporting Answer

Kadets–Shepelska–Werner arXiv:0806.1815 explicitly identifies this question in its introduction, saying it was reiterated in Shvidkoy's paper and in Kadets–Kalton–Werner. The paper introduces poor subspaces and proves, in Theorem 6.10 and the following corollary in the arXiv/published numbering, that there is a subspace

$$E \subset L_1[0,1]$$

which is isomorphic to ℓ_1, and therefore has the Radon–Nikodym property, while

$$L_1[0,1]/E$$

fails the Daugavet property. This is exactly a negative answer to the source question.

Included Partial History

Before the supporting counterexample was identified, this run produced a positive structured subcase, now stored inside this packet at

`included_partial_history/0210287_partition_ell1_rnp_quotient_daugavet/`.

It proves that if (A_n) is a countable measurable partition of a nonatomic probability space and

$$X = \overline{\mathrm{span}}\{\mathbf{1}_{A_n}/\lambda(A_n) : n \in \mathbb{N}\} \subset L_1,$$

then X is isometric to ℓ_1, hence has the Schur property and the Radon–Nikodym property, and L_1/X has the Daugavet property.

This positive subcase is useful context but is subsumed by the later negative literature answer. The two results are compatible: partition-generated copies of ℓ_1 can have Daugavet quotients, while arXiv:0806.1815 constructs a different copy of ℓ_1 whose quotient fails the Daugavet property.

Scope

This packet fully settles the second final question from arXiv:math/0210287 as an already-known literature answer. It does not address the separate first question asking whether there exists a rich subspace of L_1 with the Schur property.

The included partition subcase should be read as provenance and contrast, not as a separate registry result for the original question.

Search Evidence

The decisive source was found by searching the local run indexes and web for phrases including `poor subspace`, `L_1/X Daugavet RNP`, `Radon-Nikodym subspace of L_1 Daugavet property`, and the supporting title. The arXiv abstract of 0806.1815 states that a quotient of $L_1[0,1]$ over an ℓ_1-subspace can fail the Daugavet property and that this answers Pelczynski's question negatively. The source text further names arXiv:math/0210287 as one place where the question was reiterated. The prior local partial packet was moved inside this packet so the run has one canonical package for the source question.

References

[1] V. Kadets, N. Kalton, and D. Werner, *Remarks on rich subspaces of Banach spaces*, arXiv:math/0210287.

[2] V. Kadets, V. Shepelska, and D. Werner, *Quotients of Banach spaces with the Daugavet property*, arXiv:0806.1815; Bull. Pol. Acad. Sci. 56 no. 2 (2008), 131–147.

[3] Run fa_banach_001, agent_lane_13, *A Partition ℓ_1 Subcase of the L_1/X Daugavet Question*, included in this packet under `included_partial_history/0210287_partition_ell1_rnp_quotient_daugavet/`.